

Sustainable Software Engineering - RolePlay Assignment: Detecting and Refactoring Community Smells



Sustainable Software Engineering: Assignment Overview

You are members of a development team working on a complex, open-source machine learning project. Each of you plays a critical role—developer, maintainer, reviewer, or designer. The project is evolving rapidly, and collaboration is essential for success. Your team is facing a hidden challenge: a *community smell*—a recurring pattern of unhealthy communication or collaboration behavior that is slowing you down and causing friction.

Your mission:

- Identify the dominant community smell/s in your team's scenario (i.e., organizational silos, lone wolf, radio silence or black cloud), additionally others in the tables.
- Analyze communication logs, team structures, and metrics provided.
- Analyze possible consequences scenario.
- Propose concrete sociotechnical refactoring strategies to fix the issues.
- Prepare 5 minutes presentation.

Remember:

This is not about fixing code—it's about *refactoring the team's social architecture* to ensure long-term collaboration health.

Sustainable Software Engineering: Assignment Overview

Time	Activity	Description
13:30 – 13:40	Introduction	Overview of the assignment
13:40 – 13:50	Team Formation & Scenario Distribution	Students are divided into groups, each receiving one project scenario.
13:50 – 14:20	Group Analysis and Discussion	Teams analyze their scenario (communication logs, metrics, diagrams).
14:20 – 14:30	BREAK	
14:30 – 15:10	Group Presentations	Each team presents their findings and refactoring strategies (5 mins each).
15:10 – 15:25	Feedback & Open Discussion	Instructor highlights key points, asks questions, opens the floor to peers.
15:25 – 15:30	Wrap-up	Learning outcomes

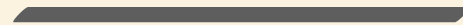
ORGANIZATIONAL SILO



BLACK CLOUD



LONE WOLF



RADIO SILENCE



Community Smells



ORGANIZATIONAL SILO

This smell occurs when there are community siloed areas that do not communicate with each other, except through one or two of their respective members.

Community Smells



BLACK CLOUD

This smell appears when there is an excessive information overload due to the lack of structured communication.

Community Smells



LONE WOLF

The presence of this smell occurs when there are contributors who work in an irrespectful manner or regardless of their team.

Community Smells



RADIO SILENCE

This smell occurs when a high degree of formality in regular procedures or a specific structural organization inefficiently channels all communication through one individual or a very small group of people (a "bottleneck" or "unique boundary spanner")

Community Smells



REFACTORING STRATEGIES

STRATEGY	DEFINITION
Restructure the community	Organizing sub-teams heterogeneously, splitting larger teams in smaller ones, swapping team members every task/story/sprint.
Create communication plan	Creating appropriate communication channels and protocols.
Mentoring	Talking with the team members, trying to solve communication problems.
Cohesion exercising	Doing exercises in order to improve the cohesion of the team (e.g., brainstorming, number of meetings).
Monitoring	Monitoring closely how team members communicate and share information.
Introduce a social-rewarding mechanism	Providing benefits to team members who work and communicate ap- propriately. (e.g., credits)
Not relevant	Opinions not related to refactoring strategies.

ORGANIZATIONAL SILO

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Introduce a social sanctioning mechanism	Motivating people to act as expected.

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Sustainable Software Engineering: Taxonomy of Community Smells

Community Smell	Description
Architecture by osmosis	Architecture is undocumented and shared informally.
Architecture hood	Architects are disconnected or invisible to the team.
Black cloud	Critical knowledge concentrated among a few → negative influence.
Class cognition	Code structure is hard to understand, causing cognitive overload.
Code red	Communication breaks down during critical issues.
Cognitive distance	Members think differently due to role, space, or culture, causing friction.
Cookbook development	Teams mechanically follow instructions without real comprehension.
DevOps clash	Conflict between development and operations teams.
Disengagement	Contributors withdraw or lose motivation.
Dispersion	Geographic or timezone distribution causes communication failures.
Dissensus	Frequent disagreements prevent consensus.
Hyper community	Too much communication creates noise and stress.
Informality excess	Overreliance on informal communication leads to chaotic processes.
Institutional isomorphism	Blindly copying processes from other teams or companies.
Invisible architecting	Design decisions happen without involving key collaborators.

Sustainable Software Engineering: Taxonomy of Community Smells

Community Smell	Description
Leftover techie	Senior developer avoids teaching or collaboration.
Lone wolf	A developer isolates themselves, not sharing work or knowledge.
Lonesome architecting	Architect works alone, detached from development processes.
Newbie free-riding	Newcomers rely too heavily on others without contributing proportionally.
Obfuscated architecting	Critical architectural knowledge is inaccessible or unclear.
Organizational silo	Team segmentation prevents cross-team communication.
Organizational skirmish	Conflicts or rivalries between groups or roles.
Power distance	Team hierarchy prevents equal participation in decision-making.
Priggish members	Elitist team members discourage others from participating.
Prima donnas	Contributors prioritize personal goals over collaboration.
Radio silence / Bottleneck	No response to messages or requests; progress is blocked.
Sharing villainy	Knowledge is hoarded for personal advantage.
Solution defiance	Developers reject agreed-upon solutions or instructions.
Time warp	Misaligned expectations or time usage across team members.
Unlearning	Resistance to updating old habits and practices.

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